

Correlation- Examples of variance

Eg 1. Find in terms of σ_X^2 and σ_Y^2 and $\text{Cov}(X, Y)$:
 $\text{Cov}(2X + 3Y, X - 2Y)$,

First create Cov matrix

$$\begin{matrix} & \begin{matrix} X & -2Y \end{matrix} \\ \begin{matrix} 2X \\ 3Y \end{matrix} & \begin{bmatrix} & \\ & \end{bmatrix} \end{matrix} \Rightarrow \begin{bmatrix} 2\text{Cov}(X, X) & -4\text{Cov}(X, Y) \\ 3\text{Cov}(Y, X) & -6\text{Cov}(Y, Y) \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2\text{Var}(X) & -4\text{Cov}(X, Y) \\ 3\text{Cov}(Y, X) & -6\text{Var}(Y) \end{bmatrix} \begin{matrix} \text{add} \\ = \\ \text{then} \\ \text{all} \end{matrix} \Rightarrow$$

$$\text{Cov}(2X + 3Y, X - 2Y) = 2\text{Var}(X) - \text{Cov}(X, Y) - 6\text{Var}(Y)$$

Eg 1. Find in terms of σ_X^2 and σ_Y^2 and $\text{Cov}(X, Y)$:

$$\text{Var}(X - 2Y) = \text{Cov}(X - 2Y, X - 2Y)$$

$$\begin{matrix} X & -2Y \\ \begin{bmatrix} \text{Cov}(X, X) & -2\text{Cov}(X, Y) \\ -2\text{Cov}(Y, X) & 4\text{Cov}(Y, Y) \end{bmatrix} \end{matrix} \Rightarrow \begin{bmatrix} \text{Var}(X) & -2\text{Cov}(X, Y) \\ -2\text{Cov}(Y, X) & 4\text{Var}(Y) \end{bmatrix}$$

$$\Rightarrow \text{Var}(X - 2Y) = \text{Var}(X) - 4\text{Cov}(Y, X) + 4\text{Var}(Y)$$

Definition (Correlation)

The correlation $\rho_{X,Y}$ between r.v.s X and Y is defined as

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

or

$$\rho(X, Y) = \text{Cov}\left(\frac{X - \mu_X}{\sigma_X}, \frac{Y - \mu_Y}{\sigma_Y}\right)$$

(This is undefined in the degenerate cases $\sigma_X = 0$ or $\sigma_Y = 0$.) Correlation X and Y is sometimes denoted by $\text{Corr}(X, Y)$.

Note: $-1 \leq \rho_{XY} \leq 1$.

Note: ρ_{XY} is either +1 or -1 if and only if X and Y are linear functions of one another.

Eg. Show that $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X,Y)$

$$\text{Var}(X+Y) = \text{Cov}(X+Y, X+Y)$$

$$= \text{Cov}(X,X) + \text{Cov}(X,Y) + \text{Cov}(Y,X) + \text{Cov}(Y,Y)$$

$$= \text{Var}(X) + 2\text{Cov}(X,Y) + \text{Var}(Y)$$

Proposition: If X and Y are independent then,

(i) $E[XY] = E[X] E[Y]$

(ii) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Note: Be careful, if two random variables are not independent, then the above may or not be correct. That is, if the above rules were true it does not mean that they are independent. The only thing we know if they are independent then the above is true. If (i) and (ii) are true, we only learn that X and Y are uncorrelated.

part (i) we only prove it for discrete case
and the proof is similar for continuous case
as well.

$$E[XY] = \sum_x \sum_y xy \, p_{X,Y}(x,y)$$

$$\stackrel{f}{=} \sum_x \sum_y xy \, p_X(x) p_Y(y)$$

Since X, Y are independent

$$\Rightarrow E[XY] = \underbrace{\sum_x x p_X(x)}_{E[X]} \underbrace{\sum_y y p_Y(y)}_{E[Y]} = E[X] E[Y]$$

(ii) since X, Y are independent, so

$$E[XY] = E[X]E[Y] \Rightarrow \text{Cov}(X,Y) = E[XY] - E[X]E[Y] = 0$$

$$\text{Var}(X+Y) = \text{Cov}(X+Y, X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y)$$

$$\begin{aligned} \text{Cov}(X,Y) &= 0 \\ \Rightarrow \end{aligned}$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

Eg. A factory produces high-quality glassware. When the furnace temperature is correctly maintained, the glass defect rate is 5%. However, there is a 10% chance that the temperature control system malfunctions, in which case the defect rate rises to 30%.

A random sample of 20 glass items was inspected, and 4 of them were found to be defective. What is the probability that the furnace temperature control had malfunctioned?

M = the system has a malfunction

D = the glass has a defect

$$P(D | M^c) = 0.05$$

$$P(D | M) = 0.3$$

$$P(M) = 0.1$$

X = number of defective glasses

$$P(M | X=4) = \frac{P(X=4 | M) P(M)}{P(X=4)}$$

$$X \sim \text{Bin}(n=20, p = P(D))$$

$$X|M \sim \text{Bin}(n=20, p = P(D|M))$$

We should calculate $P(D)$

$$P(D) = P(D|M) P(M) + P(D|M^c) P(M^c)$$

$$P(D) = (0.3)(0.1) + (0.05)(0.9)$$

$$P(D) = 0.03 + 0.045 = 0.075$$

$$P(M | X=4) = \frac{\binom{20}{4} (0.3)^4 (0.7)^{16} (0.1)}{\binom{20}{4} (0.075)^4 (1 - 0.075)^{16}}$$

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$$= \frac{(0.3)^4 (0.7)^{16} (0.1)}{(0.075)^4 (0.925)^{16}}$$

$$= \left(\frac{300}{75}\right)^4 \left(\frac{700}{925}\right)^{16} \frac{1}{10}$$